Upward scale shift in environmental struggles. A regression analysis of the relation between performances and challengers, Peru 2001-2020

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Abstract

This paper resumes the analysis of contentious politics in Peru's post-transition period by exploring the relation between the number of challengers and the number of performances in a concrete field of contention characterized by the execution of obstructive struggle performances for the revindication of environmental claims. Indeed, as the number of challengers is an indicator of a higher level of coordination and upward scale shift, this paper explores the consistency of relying on the relation between number of performances and number of challengers as a proxy measure to trace the scale shift process in concrete fields of contention by accounting for the effect of the number of performances over the number of challengers on time. The size of the sample considered for the multivariate regression analysis is 11,543 protests registered in the BPSP from 2001 to 2020 from a total population of 21,921 protests registered from 1980 to 2020. As I will show in the results section, the regression shows a significant and increasing effect of the number of performances and year over the number of challengers over time. Within a 11,543 protest sample the proportion of variability defined by the R-square is 0.689, so the regression model explains the 6.89% of the variance of number of challengers. This multivariate regression demonstrates that the relationship is very significant at a p-value < 0.001 in determining the number of challengers in a one-day protest. The regression statistics support the following hypothesis: the number of performances as a variable measuring the count of different performances of struggle is correlated positively and is a significant predictor relative to the number of challengers as a measure of the level of coordination for the execution of obstructive environmental protests as a dimension of the process of scale shift in the environmental Peruvian field of contention from 2001-2020.

1. Introduction

The upward-scale shift of contentious events is usually associated with the coordination among different challengers for the revindication of a common claim. The escalation of protests, indeed, implies not only a higher number of individual participants (this number is rarely captured in the catalogs of protest events built for social research); instead, the number of collective actors identified with a label for which access to political decision-making has to be pursued by non-institutional means such as protest (defined as challengers), is an indicator of the level of coordination involved in the preparation of the performances of struggle and its execution.

Following the theory of contentious politics scale shift process, oriented by the assumption that the higher the number of challengers, the higher the level of coordination for a protest necessary for its escalation; I will analyze the quantitative relation between performances and challengers. But also, the upward scale of a protest due to the participation of different challengers that involves the execution of different performances of struggle in a one-day protest.

These assumptions oriented theoretically the following multivariate regression analysis, by exploring its consistency in a concrete field of contention. Indeed, the research question for this paper is: *what is the effect of time and the number of performances over the number of challengers involved in the execution of environmental obstructive struggles in Peru from 2001 to 2020?*

The following hypothesis is stated: *the number of performances as a variable measuring the count of different performances of struggle is correlated positively and is a significant*

predictor relative to the number of challengers as a measure of the level of coordination for the execution of obstructive environmental protests in the Peruvian contention from 2001-2020.

As can be evidenced by the multiple linear regression considering as dependent or outcome variable the number of challengers involved in a daily base protest, the number of performances executed in each protest, and the reference categories obstructive performances and environmental claims as values for their respective variables, and the year as an ordinal variable.

2. Data and methods

The analysis regarding the measure of the escalation of environmental protests executing obstructive performances from 2001 to 2020, considers for the application of a multivariate linear regression the following variables: number of challengers (act_n_6), number of performances (perf_n), type of performances (perf_id), type of claims (dem_id), and year (y).

2.1. Sample and population

The unit of analysis of Protest Event Analysis (PEA) (Koopmans and Rucht 2002) and the construction of contentious event catalogs (Tilly 2002) is defined by the execution of a performance of struggle identified with a political claim made by more than 10 participants on a day. This last specification is important because if a protest endures more than one day, it will count again in the database for the regression analysis. So, the registration of certain features of those events is mainly relative to who protests against whom, how, and why

(Wada 2004), and the codes and variables applied for analysis are measuring those features observed for a one-day protest event.

The Peruvian Protests Database (BPSP are its Spanish acronyms) was developed mainly under the direction of Moises Arce (2014a; 2014b) codifying 21,921 protest events reported in the three main national newspapers (El Comercio, La República, and Expreso) since January 1st 1980, and December 31st 2020 based in the revision of each daily edition of those sources in the archives. The size of the sample considered for the multivariate regression analysis is 11,543 protests registered in the BPSP from 2001 to 2020.

2.2. Variables definition and descriptive statistics

Table 1 provides the descriptive statistics for the five variables used in the multivariate regression analysis. The variables used in this regression analysis measure the array of challengers engaging in obstructive performances with particular claims on environmental issues between 2001 and 2020. This analysis speaks to the level of coordination of challengers required in an escalation of performances within one-day protests. At this point, it is important to define the variables in a manner in which its measurement could make sense for the analysis and interpretation of political contention using multivariate linear regression as a quantitative technique for inference.

Firstly, the dependent variable act_n_6 measures the "number of challengers in a protest", defining challengers as collective actors without access to formal political decision-making bodies. This number of collective actors participating in a protest is collected from daily newspapers that mentioned or made records of the protest events and coinciding political claims. Within the 11,543 cases observed, the mean for the "number of challengers per

protest" between 2001 and 2020 is 1.296 with a standard deviation of .66. This indicates that there is at least one challenger for every protest recorded. Within three standard deviations from the mean, the range for "number of challengers in a protest" exists between -.66 and 3.276 (three challengers)

The variable perf_n measures the number of performances executed within one day. This independent variable measures the count of performances that occur within "one-day protests". Noted, these one-day protest events imply the demonstration of several performances such as riots, marches, and/or road blockades, among other protest actions. A higher number of performances executed within these one-day protests is indicative of an escalation of the contention made possible with higher levels of coordination needed for the execution of multiple performances. The mean for the variable perf_n is 1.374, or that on average we are likely to see between one and two performances executed within an observed "one-day" protest, with a standard deviation of 0.63, or about half, considering a range between -.516 to 3.264 (three performances).

The variable perfor_id measures the type of performances executed, with categories of "obstructive performances", "disruptive performances", "radical performances", and "other". The grouping for the variable's categories is based on the recategorization of initial values in the original variable perf_1_code. I based the grouping on whether the performances on the following criteria: the potential to obstruct capital circulation in a national economy base (obstructive performances) and its potential to disrupt the rate of profit or rate of surplus value of a particular branch of industry or corporation (disruptive performances), its potential to radically demand the satisfaction of protest claims by struggle performances which also involve higher levels of repression (radical performances), and a

category of others in which the lowest frequencies were aggregated. The mean and standard deviation of this variable are not meaningful for the analysis of contentious politics, because it is measured as a categorical nominal variable.

The variable dem_id measures the type of claim in regard of which the protest was being executed, this implies eight categories of labor, environmental, territorial, water energy, and health access, infrastructure, public managerial decisions, national budget, and other claims. The interpretation of the mean and standard deviation couldn't be considered for analysis because is a nominal categorical variable.

Finally, the year variable y is defined as the time variable, observed between 2001 to 2020. This period captures the years in which the expansion of the extractive frontier started with the foreign direct investment not only from the US, but from China, Canada, and Japan, in which I focus on observing its effect on the dependent variable act_n_6 which is measuring our outcome of interest: the higher level of coordination between different challengers executing a protest, observed by the higher number of challengers counted in the variable act_n_6, is predicted significantly by the number of performances (perf_n).

Table 1. Descriptive statistics of number of challengers

Variable	Observations	Mean	Std. dev.	Min	Max
Count of challengers	11,547	1.296094	0.6607355	1	6
Count of performances	11,549	1.374578	0.6376628	0	4
Category of performances					
disruptive	11,549	0.4040177	0.4907222	0	1
radical	11,549	0.0757641	0.2646318	0	1
other performances	11,549	0.0357607	0.1857009	0	1
Kind of claim					
labor	11,545	0.3369424	0.4726855	0	1
territorial	11,545	0.0480728	0.2139293	0	1
water, energy, health	11,545	0.1073192	0.309532	0	1
infrastructure	11,545	0.1998268	0.3998873	0	1
public management decisions	11,545	0.2200087	0.4142701	0	1
national budget	11,545	0.0091815	0.0953832	0	1
other claims	11,545	0.0107406	0.1030832	0	1
year	11,549	2009.321	5.123307	2001	2020

Source: BPSP (Arce 2014). Filter of cases considering the period 2001-2020.

3. Results

For Table 2, the multivariate regression models the number of performances and year (between 2001 to 2020) in relation with the number of challengers, contrasting for the reference categories of obstructive protests for the variable perfor_id and environmental claims for dem_id. The regression shows a significant and increasing correlation over time. Within a 11,543 protest sample size the proportion of variability defined by the R-square is 0.689, so the regression model explains the 6.89% of the variance of number of challengers. The coefficient of 0.689 R-squared could be lower also because of the large size of the sample (Table 3). In analyzing the nominal variables, the reference categories are omitted from the coefficient table, because the effects of being in each category are measured relative to the

omitted reference category. I am using the category "obstructive performances" as the reference category for the variable perf_id to determine the likelihood that the main performance is obstructive or not. As for the variable dem_id, I am using the category "environmental claims" as the reference category to predict the likelihood that the reason for the protest is based on environmental claims. In this way, I can assess the effects of different performance types (perf_id) and different claim types (dem_id) on the number of challengers for a particular one-day protest (act_n_6).

When looking at the full regression model, we see that with one unit increase of challengers in a one-day protest (act_n_6), there is a positive association to the number of performances (perf_n). In contrast to the reference category of "obstructive performances" for the variable perfor_id, there is a positive association to the category of "disruptive performances, a negative association with the category of "radical performances", and a negative association with the category of "other" performances. In contrast to the reference category of "environmental claim" for the variable "dem_id", there is a negative association to the "labor" claims, positive association to "water, energy, and health access" claims, positive association to "public managerial decisions" claims, positive association to "national budget" claims, and positive association to "other claims"; and lastly a positive association to year "y" between 2001 and 2020.

When accounting for the association of ordinal variables to the dependent variable measuring the number of challengers, the regression shows that there is a positive association to the number of performances (perf_n) at 15% increase and 1.5% yearly increment in the number of challengers between 2001 to 2020. For types of performances (perfor_id): "disruptive performances" are .117 more likely, "radical performances" are .096 less likely, and "other"

performances are .083 less likely to occur than obstructive performances (reference category) with every one unit increase of challengers at one-day protest. For kind of claims (dem_id): "labor issues" are 0.9 less likely to occur, "water, energy and health access" are .166 more likely to occur, "public managerial decisions" are .06 more likely to occur, "national budget" is .166 more likely to occur, and "other claim" is .17 more likely to occur than environmental claims with one unit increase on the number challengers one-day protest.

This multivariate regression demonstrates that the relationship is very significant at a p-value < 0.001 in determining the number of challengers in a one-day protest. The statistics model strengthens the idea that the number of challengers is a good indicator of the level of coordination and escalation of a protest. This is logically explained with the understanding that the more we see performances executed between 2001 and 2020, the more we will observe a higher number of challengers joining these protests. There is somewhat of a slight cumulative effect over time.

Table 2. Multivariate regression of number of performances, and year over the number of challengers at environmental obstructive protests, Peru 2001-2020

Count of Challengers	*Coefficient*	*Std. err.*	*t*	*P>t*	*[95% conf.*	*interval]*
Count of performances	0.15623	0.0099993	15.62	0	0.1366296	0.1758304
Kind of performances						
disruptive performances	0.1176088	0.0137041	8.58	0	0.0907464	0.1444713
radical performances	-0.0961816	0.0241443	-3.98	0	-0.1435085	-0.0488548
other	-0.083278	0.0334594	-2.49	0	-0.148864	-0.0176919
dem_id						
labor	-0.091653	0.0250421	-3.66	0	-0.1407399	-0.0425662
territorial	-0.0460108	0.0358574	-1.28	0.2	-0.1162974	0.0242757
water, energy and health access	0.1664808	0.0292709	5.69	0	0.1091048	0.2238568
infrastructure	0.016316	0.0266669	0.61	0.5	-0.0359557	0.0685876
public managerial decisions	0.05857	0.0273309	2.14	0	0.0049969	0.1121432
national budget	0.1666511	0.0678302	2.46	0	0.0336924	0.2996098
other claims	0.1731774	0.0635139	2.73	0	0.0486794	0.2976753
y	0.015231	0.0013287	11.46	0	0.0126265	0.0178355
_cons	-29.56407	2.665806	-11.09	0	-34.7895	-24.33864

Source: BPSP (Arce 2014). Filter of cases considering the period 2001-2020.

Table 3. Analysis of variance for the multivariate regression of number of performances, and year over the number of challengers at environmental obstructive protests, Peru 2001-2020

Source	SS	df	MS
Model	347.275498	12	28.9396248
Residual	4692.61985	11,530	0.40699218
Total	5039.89535	11,542	0.43665702

Number of observations	11,543
F (12, 11530)	71.11
Prob > F	0
R-squared	0.0689
Adj R-squared	0.0679
Root MSE	0.63796

Source: BPSP (Arce 2014). Filter of cases considering the period 2001-2020.

4. Conclusion

Within the theory of contentious politics and social movement studies, a higher number of challengers would require a higher level of coordination for collective action (Tilly and Tarrow 2015). But also, the upward scale of a protest due to the participation of different challengers involves the execution of different performances of struggle in a single daily protest. These assumptions oriented theoretically the following multivariate regression analysis, by exploring its consistency in a concrete field of contention that is the social movements and protests within Peru from 2001 to 2020. The regression statistics support the following hypothesis: the number of performances as a variable measuring the count of different performances of struggle is correlated positively and is a significant predictor relative to the number of challengers as a measure of the level of coordination for the execution of obstructive environmental protests in the Peruvian contention from 2001-2020.

5. References

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